Homework for Week 12

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1.

(1) $$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2\frac{\mathrm{d}y}{\mathrm{d}x}\right)+(\lambda x+x^2)=0, p(x)=x^2, q(x)=-x^2, w(x)=x.$$

(2) $$\frac{\mathrm{d}}{\mathrm{d}x}\left[x^a(x-1)^{b-a}\frac{\mathrm{d}y}{\mathrm{d}x}\right]+\lambda x^{a-1}(x-1)^{b-a-1}y=0,$$
$$p(x)=x^a(x-1)^{b-a}, q(x)=0, w(x)=x^{a-1}(x-1)^{b-a-1}.$$

(3) $$\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{x\mathrm{e}^{-x}}{(x-1)^2}\frac{\mathrm{d}y}{\mathrm{d}x}\right]+\frac{\lambda\mathrm{e}^{-x}}{(x-1)^2}y=0, p(x)=\frac{x\mathrm{e}^{-x}}{(x-1)^2}, q(x)=0, w(x)=\frac{\mathrm{e}^{-x}}{(x-1)^2}.$$

(4) $$\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{-x^2}\frac{\mathrm{d}y}{\mathrm{d}x}\right)+2\lambda\mathrm{e}^{-x^2}=0, p(x)=\mathrm{e}^{-x^2}, q(x)=0, w(x)=2\mathrm{e}^{-x^2}.$$

2. 考虑 $y(x)\in D\left(\hat{L}\right), z(x)\in L^2[a,b]$，有

$$\left\langle z\middle|\hat{L}y\right\rangle=\left\langle\hat{L}^\dagger z\middle|y\right\rangle$$

$$\Rightarrow\left\langle z\middle|\hat{L}y\right\rangle=\int_a^b z^*\left\{-\frac{\mathrm{d}}{\mathrm{d}x}\left[p(x)\frac{\mathrm{d}y}{\mathrm{d}x}\right]+q(x)y\right\}\mathrm{d}x$$

$$=-z^*p(x)\frac{\mathrm{d}y}{\mathrm{d}x}\bigg|_a^b+\int_a^b p(x)\frac{\mathrm{d}z^*}{\mathrm{d}x}\frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x+\int_a^b q(x)z^*(x)y(x)\,\mathrm{d}x$$

$$=\left[y(x)p(x)\frac{\mathrm{d}z^*(x)}{\mathrm{d}x}-z^*(x)p(x)\frac{\mathrm{d}y(x)}{\mathrm{d}x}\right]_a^b-\int_a^b\frac{\mathrm{d}}{\mathrm{d}x}\left[p(x)\frac{\mathrm{d}z^*}{\mathrm{d}x}\right]y(x)\,\mathrm{d}x+\int_a^b q(x)z^*(x)y(x)\,\mathrm{d}x$$

$$=\left[y(x)p(x)\frac{\mathrm{d}z^*(x)}{\mathrm{d}x}-z^*(x)p(x)\frac{\mathrm{d}y(x)}{\mathrm{d}x}\right]_a^b+\left\langle\hat{L}'z\middle|y\right\rangle=\left\langle\hat{L}^\dagger z\middle|y\right\rangle.$$

若 $\hat{L}$ 的伴算符存在，即要求边界项为 0，整理并代入 $y\in D\left(\hat{L}\right)$ 满足的边界条件得到

$$y(a)\left[\frac{\cos\alpha}{c_1}z'^*(b)-\frac{\cos\beta}{c_2}z^*(b)-z'^*(a)\right]+y'(a)\left[-\frac{\sin\alpha}{c_1}z'^*(b)-\frac{\sin\beta}{c_2}z^*(b)+z^*(a)\right]=0.$$

上式对任意 $y(a), y'(a)$ 成立，故伴算符存在条件为

$$z'(a) = \frac{\cos\alpha}{c_1} z'(b) - \frac{\cos\beta}{c_2} z(b), \tag{0.1}$$
$$z(a) = \frac{\sin\alpha}{c_1} z'(b) + \frac{\sin\beta}{c_2} z(b), \tag{0.2}$$

此时 $\hat{L}$ 的伴算符为

$$\hat{L}^\dagger = -\frac{\mathrm{d}}{\mathrm{d}x}\left[p(x)\frac{\mathrm{d}}{\mathrm{d}x}\right] + q(x),$$
$$D\left(\hat{L}\right) = \{y(x) : a \le x \le b, y(x) \in L^2[a,b] \text{ 且满足 (0.1) 和 (0.2)}\},$$

当且仅当 $\sin(\alpha+\beta) = c_1 c_2$ 时 $\hat{L}$ 是自伴算符.

3. 本征函数具有形式 $\mathrm{e}^{\pm\mathrm{i}\lambda x}$，取线形组合并带入边界条件得到

$$C_1 + C_2 = 0,$$
$$(1-\beta)C_1\mathrm{e}^{\mathrm{i}\lambda l} - (1+\beta)C_2\mathrm{e}^{-\mathrm{i}\lambda l} = 0.$$

由以上两式得到

$$2\mathrm{i}\lambda l = \ln\frac{\beta+1}{\beta-1},$$

并考虑到对数函数的多值性与 $0 < \beta \le 1$，取近似得到本征值

$$2\mathrm{i}\lambda l = \ln\frac{\beta+1}{\beta-1} \approx 2\beta + \mathrm{i}(2n+1)\pi \Rightarrow \lambda_n = \frac{-2\mathrm{i}\beta + (2n+1)\pi}{2l}, n \in \mathbb{Z}.$$

相应的本征函数为 $\{\mathrm{e}^{\beta x/l}\mathrm{e}^{\mathrm{i}\frac{(2n+1)x}{2l}} - \mathrm{e}^{-\beta x/l}\mathrm{e}^{-\mathrm{i}\frac{(2n+1)x}{2l}}\}$.

4. 本征函数具有形式 $\mathrm{e}^{-\mathrm{i}\lambda x}$，带入边界条件得到

$$\mathrm{e}^{-\mathrm{i}\lambda b} = \mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\lambda a} \Rightarrow \lambda_n = \frac{\theta + 2n\pi}{a-b}, n \in \mathbb{Z},$$

相应的本征函数为 $\{\mathrm{e}^{\mathrm{i}\frac{\theta+2n\pi}{b-a}x}\}$.